

Material Safety Data Sheet (MSDS)

Facilities: Bahrain Steel

**Section 1 – Identification****1(a) Product Identifier Used on Label:** Iron Ore, Oxide Pellets**1(b) Other Means of Identification:** Iron Ore, Iron Ore Pellets**1(c) Recommended Use of the Chemical and Restrictions on Use:** Iron Ore Pellet plant, No restrictions**1(d) Name, Address, and Telephone Number:**

Bahrain Steel BSCC E.C.

Phone number : (+973) 17 464 222

Building 250, road N^o 17,

email at: info@foulath.com.bh

Hidd-117, PO Box: 50177, Dry dock highway,

Kingdom of Bahrain.

Section 2 – Hazard(s) Identification

2(a) Classification of the chemical: Iron Ore / Pellets are **not considered** hazardous under Reach regulation (REACH REGULATION (EC) No 1907/2006) and is not subject to classification under CLP regulation (REGULATION (EC) No 1272/2008). However, **Iron Ore Pellets is hazardous** under OSHA's Hazard Communication Standard (29 CFR 1910.1200). Therefore, the categories of Health Hazards as defined in "GLOBALLY HARMONIZED SYSTEM OF CLASSIFICATION AND LABELLING OF CHEMICALS (GHS), Third revised edition ST/SG/AC.10/30/Rev. 3" United Nations, New York and Geneva, 2009 have been evaluated. Refer to Section 3, 8 and 11 for additional information.

2(b) Signal Word, Hazard Statement(s), Symbols and Precautionary Statement(s):

Hazard Symbol	Hazard Classification	Signal Word	Hazard Statement(s)
	Carcinogenicity -1A	Danger	May cause cancer. May cause respiratory irritation.
	Single Target Organ Toxicity (STOT) Single Exposure -3		

Precautionary Statement(s):

Prevention	Response	Storage/Disposal
Avoid breathing dusts. Wear protective gloves / protective clothing / eye protection / face protection. Use only outdoors or in a well-ventilated area. Obtain special instructions before use. Do not handle until all safety precautions have been read and understood.	If exposed, concerned or feel unwell: Get medical advice/attention. If inhaled: Remove person to fresh air and keep comfortable for breathing. Call a poison center or doctor/physician if you feel unwell.	Dispose of contents in accordance with federal, state and local regulations. Store locked up.

2(c) Hazards Not Otherwise Classified: None Known**2(d) Unknown Acute Toxicity Statement (mixture):** None Known**Section 3 – Composition/Information on Ingredients****3(a-c) Chemical Name, Common Name (Synonyms), CAS Number and Other Identifiers, and Concentration:**

Chemical Name	CAS Number	EC Number	% weight
Iron Ore Pellets	65996-65-8	265-996-3	100%
This product is a complex mixture of iron oxides, calcium silicon oxides, calcium oxides, magnesium, calcium silicates, magnesium oxides and silicon dioxides. Listed below is a partial listing of the components that comprise this product:			
Iron Oxide (Magnetite & Hematite)	1309-37-1 1317-60-8	215-168-2 215-275-4	>92
Crystalline Silica (as Quartz)	14808-60-7	238-878-4	<1
Calcium Oxides	Varies	Varies	<5
Magnesium Oxides	Varies	Varies	<3

EC- European Community

CAS- Chemical Abstract Service

Iron Ore / Pellet

Section 4 – First-aid Measures

4(a) Description of Necessary Measures:

- ☐ **Inhalation:** If inhaled: Remove person to fresh air and keep comfortable for breathing. Call a poison center or doctor/physician if you feel unwell.
- ☐ **Eye Contact:** If exposed, concerned or feel unwell: Get medical advice/attention.
- ☐ **Skin Contact:** If exposed, concerned or feel unwell: Get medical advice/attention.
- ☐ **Ingestion:** If exposed, concerned or feel unwell: Get medical advice/attention.

4(b) Most Important Symptoms/Effects, Acute and Delayed (Chronic):

Acute Effects:

- ☐ **Inhalation:** Excessive exposure to high concentrations of dust may cause irritation to the eyes, skin and mucous membranes of the upper respiratory tract.
- ☐ **Eye:** Particles of iron or iron compounds may become imbedded in the eye. Excessive exposure to high concentrations of dust may cause irritation to the eyes.
- ☐ **Skin:** Skin contact with dusts may cause irritation, possibly leading to dermatitis. Skin contact with dusts may cause physical abrasion.
- ☐ **Ingestion:** Ingestion of dust may cause nausea and/or vomiting.

Chronic Effects:

Individuals with chronic respiratory disorders (i.e., asthma, chronic bronchitis, emphysema, etc.) may be adversely affected by any fume or airborne particulate matter exposure. Persons with pre-existing skin disorders may be more susceptible to dermatitis.

4(c) Immediate Medical Attention and Special Treatment: Treat symptomatically.

Section 5 – Fire-fighting Measures

5(a) Suitable (and unsuitable) Extinguishing Media: Use steam, water fog, dry chemical or carbon dioxide.

5(b) Specific Hazards Arising from the Chemical: It is unlikely based on the chemistry of this material, in the metal oxide forms that it will present an explosion hazard. However, High concentrations of airborne metallic fines may present an explosion hazard.

5(c) Special Protective Equipment and Precautions for Fire-fighters: Self-contained NIOSH approved respiratory protection and full protective clothing should be worn when fumes and/or smoke from fire are present. Heat and flames cause emittance of acrid smoke and fumes. Do not release runoff from fire control methods to sewers or waterways. Firefighters should wear full face-piece self-contained breathing apparatus and chemical protective clothing with thermal protection. Direct water stream will scatter and spread flames and, therefore, should not be used.

Section 6 - Accidental Release Measures

6(a) Personal Precautions, Protective Equipment and Emergency Procedures: If material is in a dry state, avoid inhalation of dust. Personnel should be protected against contact with eyes and skin. Fine, dry material should be removed by vacuuming or wet sweeping methods to prevent spreading of dust. Avoid using compressed air. Do not release into sewers or waterways.

6(b) Methods and Materials for Containment and Clean Up: Collect material in appropriate, labeled containers for recovery or disposal in accordance with federal, state, and local regulations. Follow applicable OSHA regulations (29 CFR 1910.120) and all other pertinent state and federal requirements.

Section 7 - Handling and Storage

7(a) Precautions for Safe Handling: Avoid breathing dusts. Wear protective gloves / protective clothing / eye protection / face protection. Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Emergency safety showers and eye wash stations should be present.

7(b) Conditions for Safe Storage, Including any Incompatibilities: Store away from acids and incompatible materials. If feasible, store locked up.

Section 8 - Exposure Controls / Personal Protection

8(a) Occupational Exposure Limits (OELs): The following exposure limits are offered as reference, for an experienced industrial hygienist to review.

Ingredients	OSHA PEL ¹	ACGIH TLV ²	NIOSH REL ³	IDLH ⁴
Iron Oxides	10 mg/m ³ (as iron oxide fume)	5.0 mg/m ³	5.0 mg/m ³ (as iron oxide dust and fume)	2,500 mg/m ³
Crystalline Silica (as Quartz)	(30mg/m ³)/(%SiO ₂ +2) (as total dust) (10mg/m ³)/(%SiO ₂ +2) (as respirable fraction)	0.025 mg/m ³	0.05 mg/m ³	50 mg/m ³
Calcium Oxide	5.0 mg/m ³	2.0 mg/m ³	2.0 mg/m ³	25 mg/m ³
Magnesium Oxide	15 mg/m ³	10 mg/m ³	NE	750 mg/m ³

NE - None Established

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Section 8 - Exposure Controls / Personal Protection

8(a) Occupational Exposure Limits (OELs) (continued):

1. OSHA PELs (Permissible Exposure Limits) are 8-hour TWA (time-weighted average) concentrations unless otherwise noted. A ("C") designation denotes a ceiling limit, which should not be exceeded during any part of the working exposure unless otherwise noted. An Action level (AL) is used by OSHA and NIOSH to express a health or physical hazard. They indicate the level of a harmful or toxic substance/activity, which requires medical surveillance, increased industrial hygiene monitoring, or biological monitoring. Action Levels are generally set at one half of the PEL but the actual level may vary from standard to standard. The intent is to identify a level at which the vast majority of randomly sampled exposures will be below the PEL.
2. Threshold Limit Values (TLV) established by the American Conference of Governmental Industrial Hygienists (ACGIH) are 8-hour TWA concentrations unless otherwise noted. ACGIH TLVs are for guideline purposes only and as such are not legal, regulatory limits for compliance purposes. A Short Term Exposure Limit (STEL) is defined as the maximum concentration to which workers can be exposed for a short period of time (15 minutes) for only four times throughout the day with at least one hour between exposures.
3. The National Institute for Occupational Safety and Health Recommended Exposure Limits (NIOSH-REL) - Compendium of Policy and Statements. NIOSH, Cincinnati, OH (1992). NIOSH is the federal agency designated to conduct research relative to occupational safety and health. As is the case with ACGIH TLVs, NIOSH RELs are for guideline purposes only and as such are not legal, regulatory limits for compliance purposes.
4. The "Immediately Dangerous to Life or Health air concentration values (IDLHs)" are used by NIOSH as part of the respirator selection criteria and were first developed in the mid-1970's by NIOSH. The Documentation for Immediately Dangerous to Life or Health Concentrations (IDLHs) is a compilation of the rationale and sources of information used by NIOSH during the original determination of 387 IDLHs and their subsequent review and revision in 1994.

8(b) Appropriate Engineering Controls: Local exhaust ventilation should be used to control the emission of air contaminants. General dilution ventilation may assist with the reduction of air contaminant concentrations. Emergency eye wash stations and deluge safety showers should be available in the work area.

8(c) Individual Protection Measures:

- **Respiratory Protection:** Seek professional advice prior to respirator selection and use. Follow OSHA respirator regulations (29 CFR 1910.134) and, if necessary, use only a NIOSH-approved respirator. Select respirator based on its suitability to provide adequate worker protection for given working conditions, level of airborne contamination, and presence of sufficient oxygen. Concentration in air of the various contaminants determines the extent of respiratory protection needed. Half-face, negative-pressure, air-purifying respirator equipped with P100 filter is acceptable for concentrations up to 10 times the exposure limit. Full-face, negative-pressure, air-purifying respirator equipped with P100 filter is acceptable for concentrations up to 50 times the exposure limit. Protection by air-purifying negative-pressure and powered air respirators is limited. Use a positive-pressure-demand, full-face, supplied air respirator or self-contained breathing apparatus (SCBA) for concentrations above 50 times the exposure limit. If exposure is above the IDLH (Immediately dangerous to life or health) for any of the constituents, or there is a possibility of an uncontrolled release or exposure levels are unknown, then use a positive-demand, full-face, supplied air respirator with escape bottle or SCBA.

Warning! Air-purifying respirators both negative-pressure, and powered-air do not protect workers in oxygen-deficient atmospheres.

- **Eyes:** Wear appropriate eye protection to prevent eye contact. Use safety glasses with side shields or chemical goggles.
- **Skin:** Persons handling this product should wear appropriate clothing to prevent skin contact. Wear protective gloves.
- **Other Protective Equipment:** An eyewash fountain and deluge shower should be readily available in the work area.

Section 9 - Physical and Chemical Properties

9(a) Appearance (physical state, color, etc.): Solid. Grayish, grayish-black, small pebbles

9(b) Odor: Odorless

9(c) Odor Threshold: NA

9(d) pH: NA

9(e) Melting Point/Freezing Point: >1,315.6 °C

9(f) Initial Boiling Point and Boiling Range: NA

9(g) Flash Point: NA

9(h) Evaporation Rate: NA

9(i) Flammability (solid, gas): Not flammable

NA - Not Applicable

ND - Not Determined for product as a whole

9(j) Upper/lower Flammability or Explosive Limits: NA

9(k) Vapor Pressure: NA

9(l) Vapor Density (Air = 1): NA

9(m) Relative Density: >3.5 SG

9(n) Solubility(ies): Insoluble

9(o) Partition Coefficient n-octanol/water: NA

9(p) Auto-ignition Temperature: ND

9(q) Decomposition Temperature: ND

9(r) Viscosity: ND

Section 10 - Stability and Reactivity

10(a) Reactivity: Not Determined (ND).

10(b) Chemical Stability: Iron Ore / Pellets are stable under normal storage and handling conditions.

10(c) Possibility of Hazardous Reaction: None Known.

10(d) Conditions to Avoid: Storage with strong acids or calcium hypochlorite.

10(e) Incompatible Materials: Iron oxide dusts in contact with calcium hypochlorite evolve oxygen and may cause an explosion.

10(f) Hazardous Decomposition Products: Toxic fumes and vapors may be released at elevated temperatures.

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Section 11 - Toxicological Information

11 Information on Toxicological Effects: The following toxicity data has been determined for **Iron Ore / Pellets** by using the information available for its components applied to the guidance on the preparation of an SDS under the GHS requirements of OSHA and the EU CPL:

Hazard Classification	Hazard Category		Hazard Symbols	Signal Word	Hazard Statement
	EU	OSHA			
Carcinogenicity (covers Categories 1A, 1B and 2)	NR	1A ^g		Danger	May cause cancer.
Specific Target Organ Toxicity (STOT) Following Single Exposure (covers Categories 1-3)	NR	3 ⁱ		Warning	May cause respiratory irritation

* NR Not Rated - Available data does not meet criteria for classification.

The Toxicological data listed below are presented regardless to classification criteria. Individual hazard classification categories where the toxicological information has met or exceeded a classification criteria threshold are listed above.

- a. No LC₅₀ or LD₅₀ has been established for **Iron Ore / Pellets**. The following data has been determined for the components:
 - ☐ **Iron Ore Pellets:** Rat LD₅₀ > 4500 mg/kg
 - ☐ **Iron Oxide:** LD₅₀ = >10,000 mg/kg (Oral/Rat)
 - ☐ **Silica:** LD₅₀ = 500 mg/kg (Oral/Rat)
- b. No Skin (Dermal) Irritation data available for **Iron Ore / Pellets** as a mixture. The following Skin (Dermal) Irritation data has been determined for the components:
 - ☐ **Iron Oxide:** Moderately irritating.
 - ☐ **Calcium Oxide:** Rabbit Not irritating (REACH).
 - ☐ **Magnesium Oxide:** Severe skin irritant in human (HSDB).
- c. No Eye Irritation data available for **Iron Ore / Pellets** as a mixture. The following Eye Irritation information was found for the components:
 - ☐ **Iron Oxide:** Severely irritating; may cause burns. Human Corrosive (IUCILID).
 - ☐ **Silicon Dioxide:** Crystalline silica may cause abrasion of the cornea.
 - ☐ **Calcium Oxide:** Rabbit Irritating (REACH).
 - ☐ **Magnesium Oxide:** Severe eye irritant in human (HSDB).
- d. No Skin (Dermal)/Respiratory Sensitization data available for **Iron Ore / Pellets** as a mixture or its individual components.
- e. No Aspiration Hazard data available for **Iron Ore / Pellets** as a mixture or its individual components.
- f. No Germ Cell Mutagenicity data available for **Iron Ore / Pellets** as a mixture. The following Germ Cell Mutagenicity information was found for the components:
 - ☐ **Iron Oxide:** Both positive and negative data.
- g. Carcinogenicity: IARC, NTP, and OSHA do not list **Iron Ore / Pellets** as carcinogens. The following Carcinogenicity information was found for the components:
 - ☐ **Iron Oxide:** IARC-3, TLV-A4.
 - ☐ **Silicon Dioxide:** Repeated exposure to crystalline silica causes lung cancer in exposed humans. IARC-1, NTP-1, TLV-A2, and OSHA.
 - ☐ **Magnesium Silicates:** Lifetime inhalation exposure of rats and mice to atmospheres of magnesium silicate resulted in tumors of the lung in female rats at $\geq 6 \text{ mg/m}^3$.
- h. No Toxic Reproduction data available for **Iron Ore / Pellets** as a mixture or its individual components.
- i. No Specific Target Organ Toxicity (STOT) following a Single Exposure data available for **Iron Ore / Pellets** as a mixture. The following STOT following a Single Exposure data was found for the components:
 - ☐ **Iron Oxide:** May cause lung irritation.
 - ☐ **Silicon Dioxide:** Single exposure to very high airborne levels may cause lung irritation in exposed humans.
 - ☐ **Calcium Oxide:** Can cause respiratory tract irritation, skin and eye irritation.
- j. No Specific Target Organ Toxicity (STOT) following Repeated Exposure data was available for **Iron Ore / Pellets** as a whole. The following STOT following Repeated Exposure data was found for the components:
 - ☐ **Iron Ore Pellets:** Rat inhalation 2 week iron oxide doses 185, 195 and 210 mg/m³ No effect level as there was an increase in all dose groups in lung weight with accumulation of test article in lungs colors black, yellow and red oxides.
 - ☐ **Iron Oxide:** Some pulmonary and lung effects reported.
 - ☐ **Silicon Dioxide:** Repeated exposure to crystalline silica causes silicosis and kidney damage as well as increased incidence of autoimmune disorders in humans.

The above toxicity information was determined from available scientific sources to illustrate the prevailing posture of the scientific community. The scientific resources includes: The American Conference of Governmental Industrial Hygienist (ACGIH) Documentation of the Threshold Limit Values (TLVs) and Biological Exposure indices (BEIs) with Other Worldwide Occupational Exposure Values 2009, The International Agency for Research on Cancer (IARC), The National Toxicology Program (NTP) updated documentation, the World Health Organization (WHO) and other available resources, the International Uniform Chemical Information Database (IUCILID), European Union Risk Assessment Report (EU-RAR), Concise International Chemical Assessment Documents (CICAD), European Union Scientific Committee for Occupational Exposure Limits (EU-SCOEL), Agency for Toxic Substances and Disease Registry (ATSDR), Hazardous Substance Data Bank (HSDB), and International Programme on Chemical Safety (IPCS).

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Section 11 - Toxicological Information (continued)

11 Information on Toxicological Effects (continued):

The following health hazard information is provided regardless to classification criteria and is based on the individual component(s):

Acute Effects by Component:

- **IRON (and Iron Oxide):** Iron is harmful if swallowed, causes skin irritation, and causes eye irritation. Contact with iron oxide has been reported to cause skin irritation and serious eye damage.
- **CRYSTALLINE SILICA (Silicon Dioxide):** Causes irritation and inflammation of the respiratory tract. May cause abrasion of the cornea. Inhalation may cause cough. A single exposure to very high airborne levels may cause lung irritation in exposed humans.
- **CALCIUM OXIDE:** Calcium oxide is an eye and skin irritant.
- **MAGNESIUM OXIDE:** Headache, cough, sweating, nausea and fever may be caused by exposure to freshly formed fumes. The symptoms of metal fume fever do not become manifest until 4-12 hours after exposure.

Delayed (chronic) Effects by Component:

- **IRON (and Iron Oxide):** Chronic inhalation of excessive concentrations of iron oxide fumes or dusts may result in the development of a benign lung disease, called siderosis which is observable as an X-ray change. No physical impairment of lung function has been associated with siderosis. Inhalation of excessive concentrations of ferric oxide may enhance the risk of lung cancer development in workers exposed to pulmonary carcinogens. Iron oxide is listed as a Group 3 (not classifiable) carcinogen by the International Agency for Research on Cancer (IARC).
- **SILICA (Crystalline Quartz):** Inhalation of quartz is classified by IARC as a probable human carcinogen. Chronic exposure can cause silicosis, a form of lung scarring that can cause shortness of breath, reduced lung function, and in severe cases, death. Repeated exposure may cause kidney damage as well as increased incidence of autoimmune disorder.
- **CALCIUM OXIDE:** Depending on the concentration and duration of exposure, repeated or prolonged inhalation may cause inflammation of the respiratory passages, ulcers of the mucous membranes, and possible perforation of the nasal septum. Repeated or prolonged skin contact may cause dermatitis.
- **MAGNESIUM OXIDE:** Irritation of eyes, nose, and throat. Symptoms may include dryness of nose and mouth, cough, feeling of weakness, tightness of chest, muscular pain, chills, fever, headache, nausea, and vomiting.

Section 12 - Ecological Information

12(a) Ecotoxicity (aquatic & terrestrial): No data available for the product, **Iron Ore / Pellets** as a whole. However, individual components of the product have been found to be toxic to the environment. Dusts may migrate into soil and groundwater and be ingested by wildlife as follows:

- **Iron Oxide:** LC₅₀: >1000 mg/L; Fish
- **Calcium Oxide:** LC₅₀: 159 mg/L; invertebrates

12(b) Persistence & Degradability: No Data Available

12(c) Bioaccumulative Potential: No Data Available

12(d) Mobility (in soil): No Data Available

12(e) Other Adverse Effects: None Known

Additional Information:

Hazard Category: No Category

Signal Word: No Signal Word

Hazard Symbol: No Hazard Symbol

Hazard Statement: No Hazard Statement

Section 13 - Disposal Considerations

Disposal: Dispose of contents/container in accordance with local/regional/international regulations.

Container Cleaning and Disposal: Follow applicable federal, state and local regulations. Observe safe handling precautions. European Waste Catalogue 10-02-99 (wastes not otherwise specified) or 16-03-04 (organic waste other than those specified).

Please note this information is for Iron Ore / Pellets in its original form. Any alterations can void this information.

Section 14 - Transport Information

14 (a-g) Transportation Information:

US Department of Transportation (DOT) under 49 CFR 172.101 **does not** regulate **Iron Ore / Pellets** as a hazardous material. All federal, state, and local laws and regulations that apply to the transport of this type of material must be adhered to.

Shipping Name: Not Applicable (NA) Shipping Symbols: NA Hazard Class: NA UN No.: NA Packing Group: NA DOT/ IMO Label: NA Special Provisions (172.102): NA	Packaging Authorizations a) Exceptions: NA b) Non-bulk: NA c) Bulk: NA	Quantity Limitations a) Passenger Aircraft or Rail: NA b) Cargo Aircraft Only: NA Vessel Stowage Location: NA DOT Reportable Quantities: NA
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International Maritime Dangerous Goods (IMDG) and the Regulations Concerning the International Carriage of Dangerous Goods by Rail (RID) classification, packaging and shipping requirements follow the US DOT Hazardous Materials Regulation.

Iron Ore / Pellet

Section 14 - Transport Information (continued)

Regulations Concerning the International Carriage of Dangerous Goods by Road (ADR) does not regulate Iron Ore / Pellets as a hazardous material.

Shipping Name: NA Classification Code: NA UN No.: NA Packing Group: NA ADR Label: NA Special Provisions: NA Limited Quantities: NA	Packaging a) Packing Instructions: NA b) Special Packing Provisions: NA c) Mixed Packing Provisions: NA	Portable Tanks & Bulk Containers a) Instructions: NA b) Special Provisions: NA
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International Air Transport Association (IATA) does not regulate Iron Ore / Pellets as a hazardous material.

Shipping Name: NA Class/Division: NA Hazard Label (s): NA UN No.: NA Packing Group: NA Excepted Quantities (EQ): NA	Passenger & Cargo Aircraft Limited Quantity (EQ) Pkg Inst: NA Max Net Qty/Pkg: NA	Cargo Aircraft Only Pkg Inst: NA Max Net Qty/Pkg: NA	Special Provisions: NA ERG Code: NA
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Pkg Inst – Packing Instructions

Max Net Qty/Pkg – Maximum Net Quantity per Package

ERG – Emergency Response Drill Code

Iron Ore / Pellets does not have a Transport Dangerous Goods (TDG) classification.

Section 15 - Regulatory Information

Regulatory Information: The following listing of regulations relating to an Bahrain Steel product may not be complete and should not be solely relied upon for all regulatory compliance responsibilities. This product and/or its constituents are subject to the following regulations:

OSHA Regulations: Air Contaminant (29 CFR 1910.1000, Table Z-1, Z-2, Z-3): The product as a whole is not listed. However, individual components of the product are listed refer to Section 8.

EPA Regulations: The product, **Iron Ore / Pellets and its components** are not listed.

SARA Potential Hazard Categories: Immediate Acute Health Hazard, delayed Chronic Health Hazard.

SARA 313 Supplier Notification: The product, **Iron Ore / Pellets** does not contain any of the toxic chemicals subject to the reporting requirements of section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 and 40 CFR part 372.

Regulations Key:

CAA	Clean Air Act (42 USC Sec. 7412; 40 CFR Part 61 [As of: 8/18/06])
CERCLA	Comprehensive Environmental Response, Compensation and Liability Act (42 USC Secs. 9601(14), 9603(a); 40 CFR Sec. 302.4, Table 302.4, Table 302.4 and App. A)
CWA	Clean Water Act (33 USC Secs. 1311; 1314(b), (c), (e), (g); 136(b), (c); 137(b), (c) [as of 8/2/06])
RCRA	Resource Conservation Recovery Act (42 USC Sec. 6921; 40 CFR Part 261 App VIII)
SARA	Superfund Amendments and Reauthorization Act of 1986 Title III Section 302 Extremely Hazardous Substances (42 USC Secs. 11023, 13106; 40 CFR Sec. 372.65) and Section 313 Toxic Chemicals (42 USC Secs. 11023, 13106; 40 CFR Sec. 372.65 [as of 6/30/05])
TSCA	Toxic Substance Control Act (15 U.S.C. s/s 2601 et seq. [1976])
SDWA	Safe Drinking Water Act (42 U.S.C. s/s 300f et seq. [1974])

State Regulations: The product, **Iron Ore / Pellets** as a whole is not listed in any state regulations. However, individual components of the product are listed in various state regulations:

Pennsylvania Right to Know: Contains regulated material in the following categories:

- Hazardous Substances: Iron oxide, Calcium oxide, Magnesium oxide, and Crystalline silica.

California Prop. 65: Contains elements known to the State of California to cause cancer or reproductive toxicity. This includes Crystalline silica (airborne particles of respirable size only).

New Jersey: Contains regulated material in the following categories:

- Hazardous Substance: Iron Oxide, Magnesium Oxide, Calcium Oxide, and Crystalline silica.
- Special Health Hazard Substances: Calcium Oxide and Crystalline silica.

Minnesota: Calcium oxide and Crystalline silica (inhaled in the form of quartz or cristobalite from occupational source).

Massachusetts: Iron Oxide (dust or fume), Calcium oxide and Crystalline silica.

Other Regulations:

WHMIS Classification (Canadian): The product, **Iron Ore / Pellets** is not listed as a whole. However individual components are listed.

Ingredients	WHMIS Classification
Calcium Oxide	E
Quartz	D-2A

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations and the SDS contains all the information required by the Controlled Products Regulations.

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Section 16 - Other Information

Prepared By: Bahrain Steel

Revision History:

6/23/2009 - Original Issue Date

8/20/2014 - Update to OSHA HAZ COM 2012

Additional Information:

Hazardous Material Identification System (HMIS) Classification

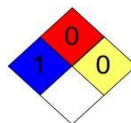
Health Hazard	1
Fire Hazard	0
Physical Hazard	0

HEALTH= 1, * Denotes possible chronic hazard if airborne dusts or fumes are generated Irritation or minor reversible injury possible.

FIRE= 0, Materials that will not burn.

PHYSICAL HAZARDS = 0, Materials that are normally stable, even under fire conditions, and will not react with water, polymerize, decompose, condense, or self-react. Non-explosives.

National Fire Protection Association (NFPA)



HEALTH = 1, Exposure could cause irritation but only minor residual injury even if no treatment is given.

FIRE = 0, Materials that will not burn.

INSTABILITY = 0, Normally stable, even under fire exposure conditions, and are not reactive with water.

ABBREVIATIONS/ACRONYMS:

ACGIH	American Conference of Governmental Industrial Hygienists	NIF	No Information Found
BEIs	Biological Exposure Indices	NIOSH	National Institute for Occupational Safety and Health
CAS	Chemical Abstracts Service	NTP	National Toxicology Program
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act	ORC	Organization Resources Counselors
CFR	Code of Federal Regulations	OSHA	Occupational Safety and Health Administration
CNS	Central Nervous System	PEL	Permissible Exposure Limit
GI, GIT	Gastro-Intestinal, Gastro-Intestinal Tract	PNOR	Particulate Not Otherwise Regulated
HMIS	Hazardous Materials Identification System	PNOC	Particulate Not Otherwise Classified
IARC	International Agency for Research on Cancer	PPE	Personal Protective Equipment
LC50	Median Lethal Concentration	ppm	parts per million
LD50	Median Lethal Dose	RCRA	Resource Conservation and Recovery Act
LD_{Lo}	Lowest Dose to have killed animals or humans	RTECS	Registry of Toxic Effects of Chemical Substances
LEL	Lower Explosive Limit	SARA	Superfund Amendment and Reauthorization Act
µg/m³	microgram per cubic meter of air	SCBA	Self-contained Breathing Apparatus
mg/m³	milligram per cubic meter of air	STEL	Short-term Exposure Limit
mppcf	million particles per cubic foot	TLV	Threshold Limit Value
SDS	Safety Data Sheet	TWA	Time-weighted Average
MSHA	Mine Safety and Health Administration	UEL	Upper Explosive Limit
NFPA	National Fire Protection Association		

Disclaimer: This information is taken from sources or based upon data believed to be reliable. Our objective in sending this information is to help you protect the health and safety of your personnel and to comply with the OSHA Hazard Communication Standard and Title III of the Superfund Amendment and Reauthorization Act of 1986. Bahrain Steel makes no warranty as to the absolute correctness, completeness, or sufficiency of any of the foregoing, or any additional, or other measures that may not be required under particular conditions. **BAHRAIN STEEL MAKES NO WARRANTIES, EXPRESS OR IMPLIED, INCLUDING THE IMPLIED WARRANTY OF MERCHANTABILITY, OR ANY IMPLIED WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE, AND ANY IMPLIED WARRANTIES OTHERWISE ARISING FROM COURSE OF DEALING OR TRADE.**

Iron Ore / Pellets

Signal Word: DANGER

Symbols:



HAZARD STATEMENTS:

May cause cancer.
May cause respiratory irritation.

PRECAUTIONARY STATEMENTS

Avoid breathing dusts.
Wear protective gloves / protective clothing / eye protection / face protection.
Use only outdoors or in a well-ventilated area.
Obtain special instructions before use.
Do not handle until all safety precautions have been read and understood.
If exposed, concerned or feel unwell: Get medical advice/attention.
If inhaled: Remove person to fresh air and keep comfortable for breathing. Call a poison center or doctor/physician if you feel unwell.
Dispose of contents in accordance with federal, state and local regulations.
Store locked up.

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